

# Worksheet on differentiation.

Note Title

2014/03/28

Determine the derivative of the following.

$$1. \quad y = x^3 + 3^x + 3^e$$

$$2. \quad y = \cos^3(4^{5x})$$

$$3. \quad x^3 y^4 + x^5 - y^6 = 1$$

$$4. \quad f(x) = \sin(e^{-2x})$$

$$5. \quad f(t) = e^{\cos \sqrt{t}}$$

$$6. \quad f(t) = \arcsin\left(\frac{1}{t}\right)$$

$$7. \quad f(x) = \arctan(3x^4)$$

$$8. \quad f(v) = \ln(1-3v)$$

$$9. \quad f(x) = \ln(2x)$$

$$10. \quad f(t) = \ln(\ln t)$$

$$11. \quad f(x) = \operatorname{cosec} \frac{\pi}{3}$$

$$12. \quad f(t) = \operatorname{cosec}(\cot(9t))$$

$$13. \quad f(x) = (x - x^3)^{10}$$

$$14. \quad f(t) = 10^{1-2t}$$

$$15. \quad f(x) = (1-2x)^{10x}$$

$$16. f(x) = \frac{1-x^3}{7+4\arcsin(5x)}$$

$$17. f(x) = \arctan(1-x) \cdot \cos(kx)$$

$$18. f(x) = \ln \left[ \frac{e^{2x} \tan 2x}{\sec^2(x^2)} \right]$$

19. Determine the equation of the tangent line to  $f(x) = \sin^2 2x$  at  $x = \frac{\pi}{6}$ .

20. At which value(s) of  $x$  can you have a tangent to the ellipse  $9x^2 + 4y^2 = 36$  which is parallel to  $y = 2x$ .

